

软件工程概论

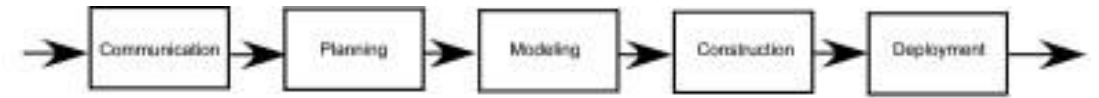
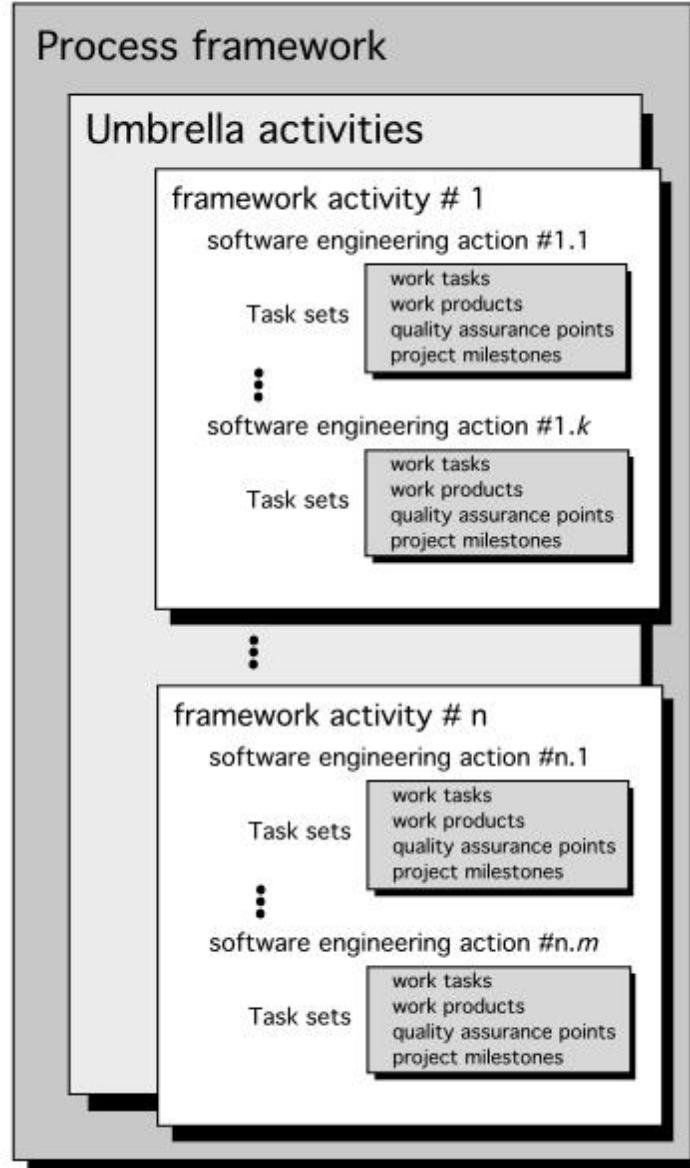
张璇 云南大学

软件过程

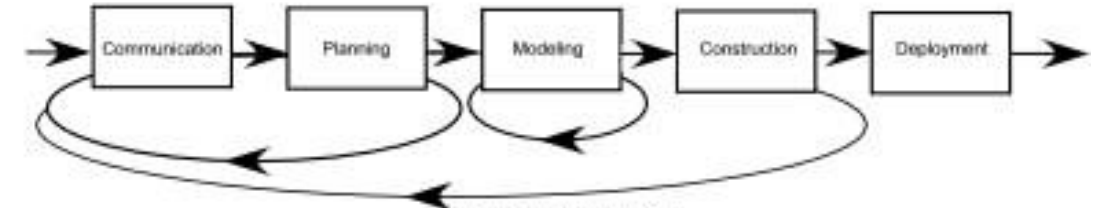
部分内容取自 *Software Engineering: A Practitioner's Approach* 作者 Roger Pressman

Software Process

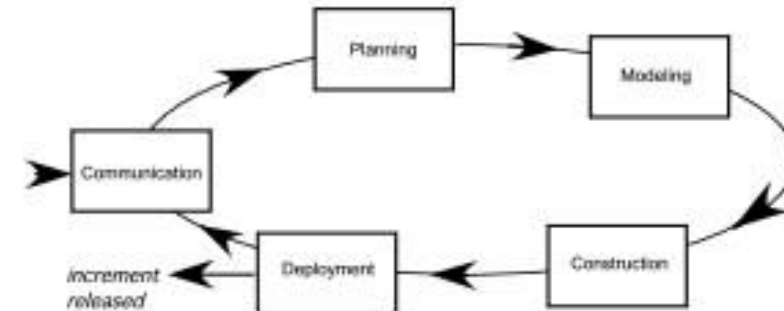
Software process



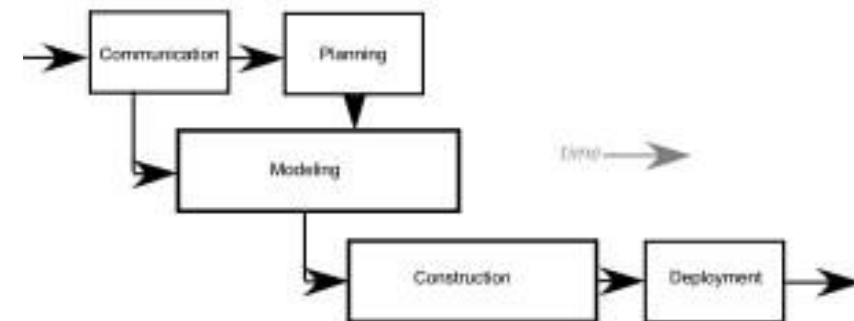
(a) linear process flow



(b) iterative process flow

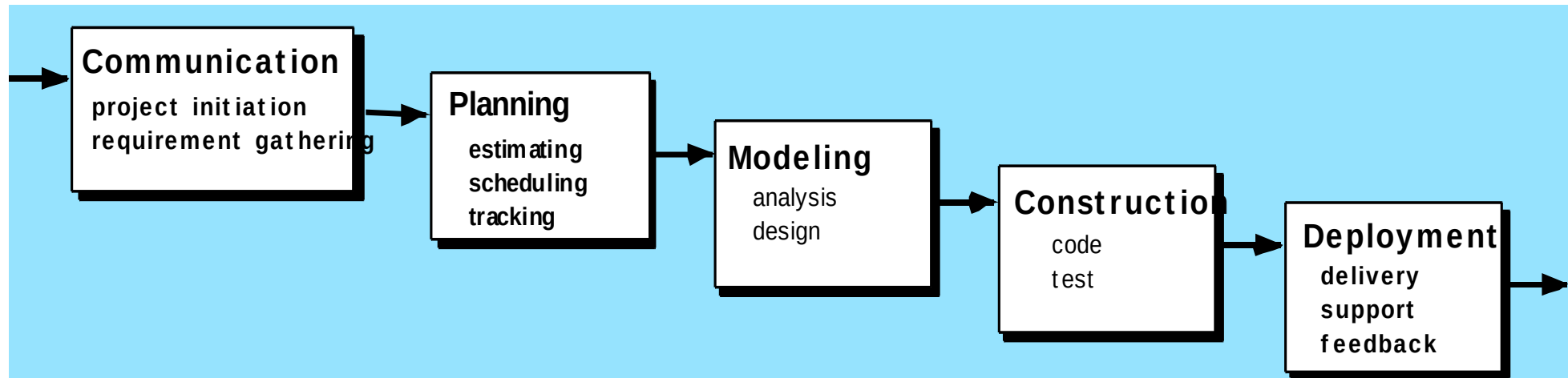


(c) evolutionary process flow

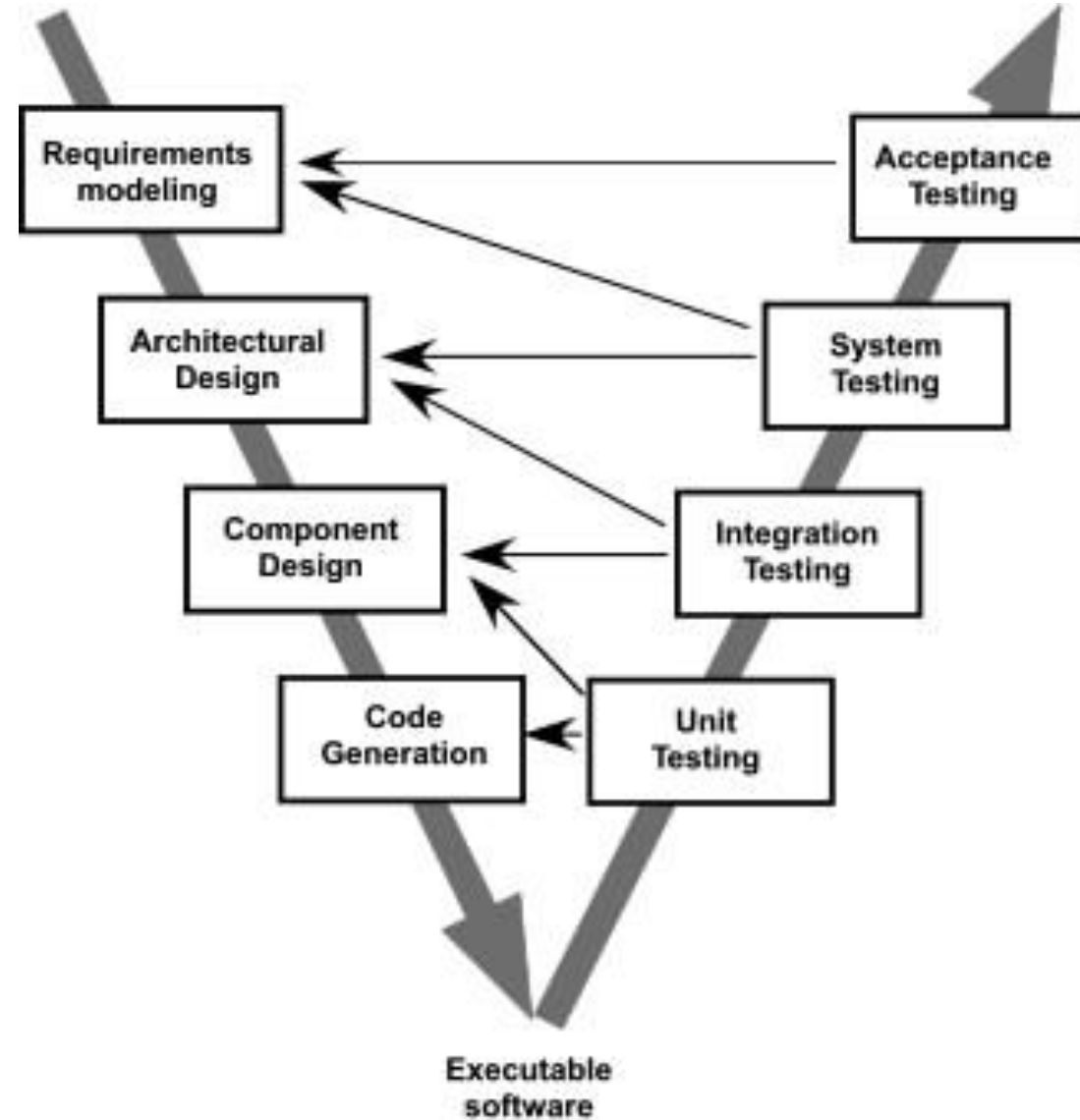


(d) parallel process flow

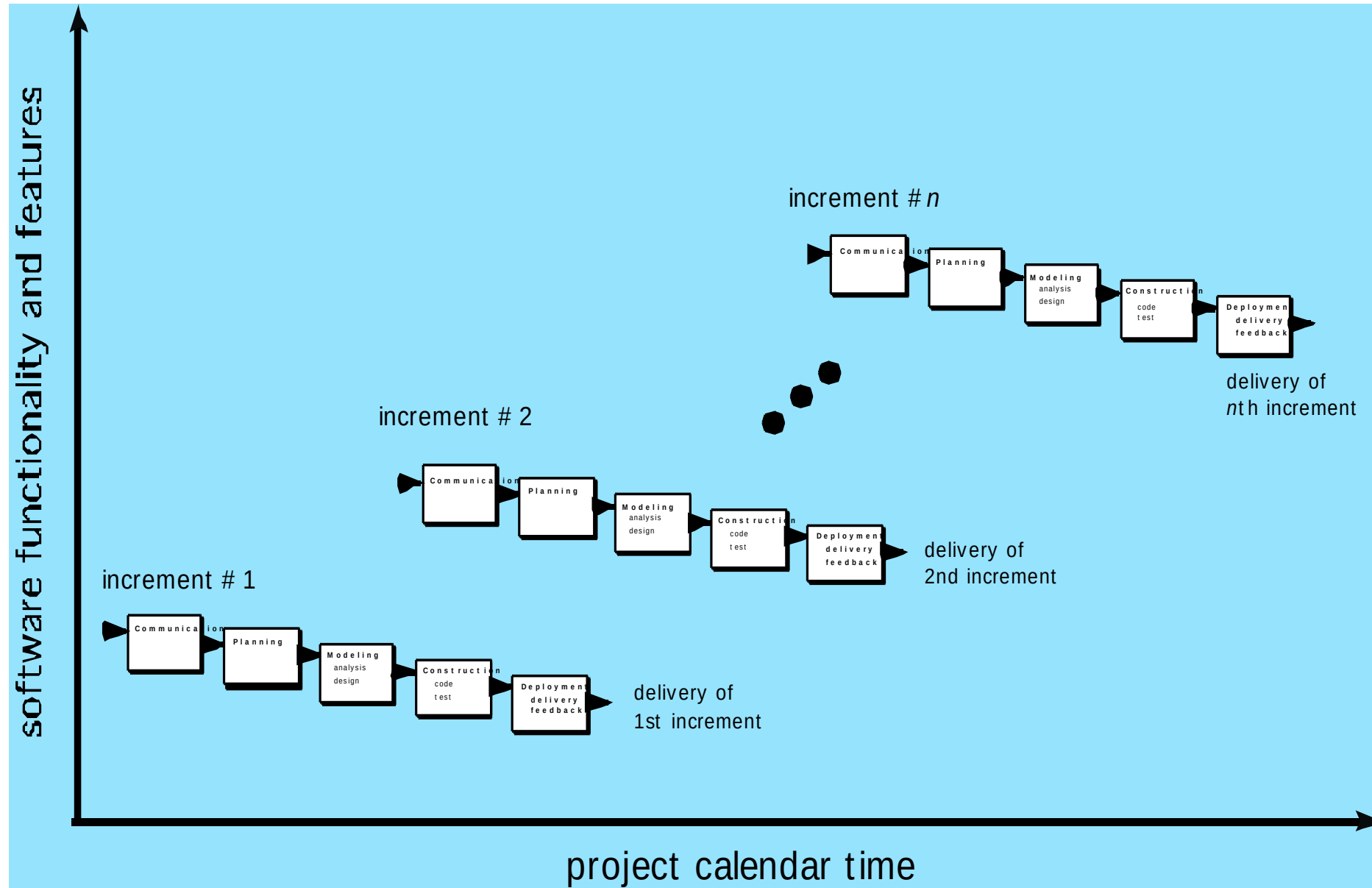
The Waterfall Model



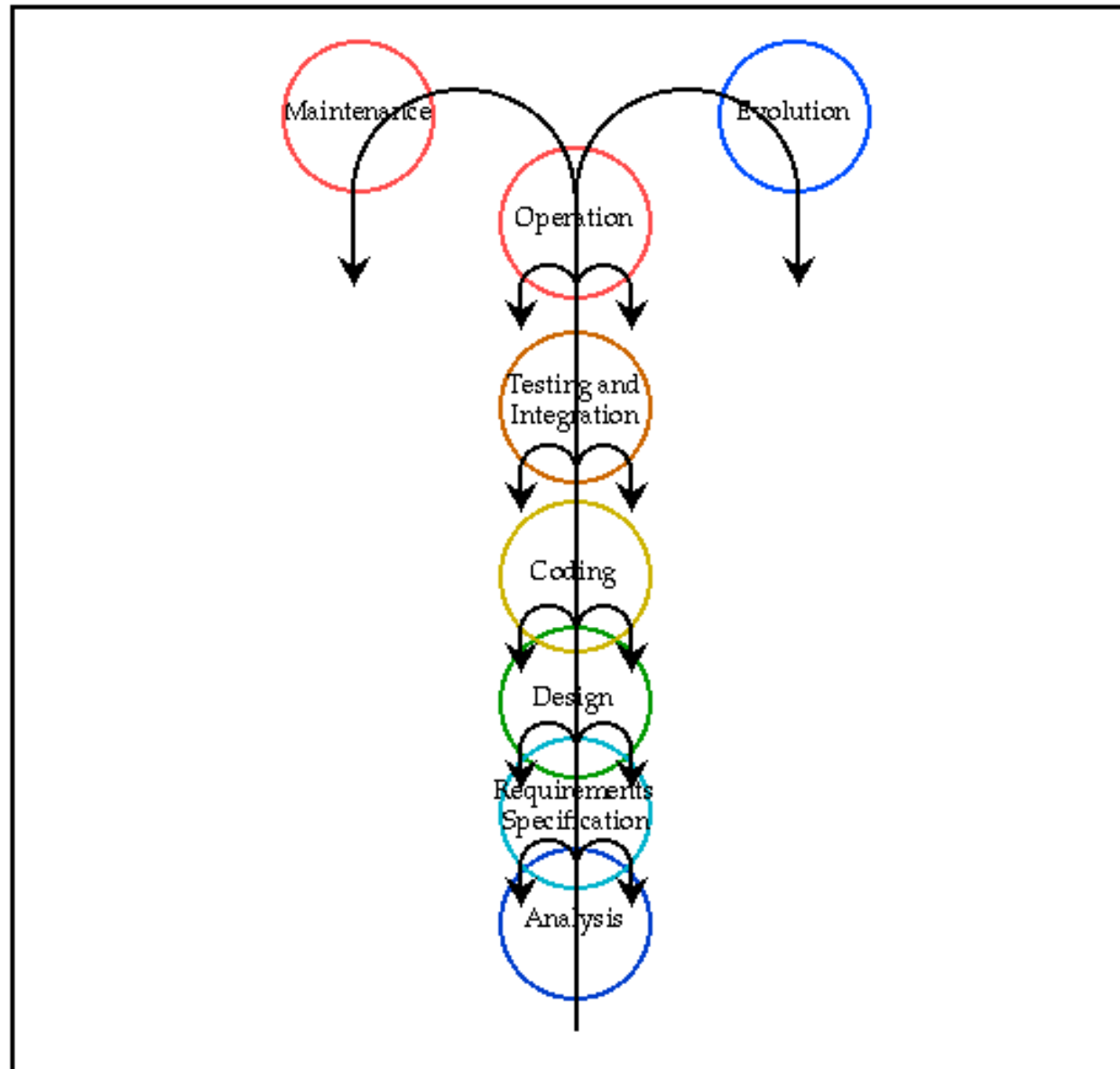
The V-Model



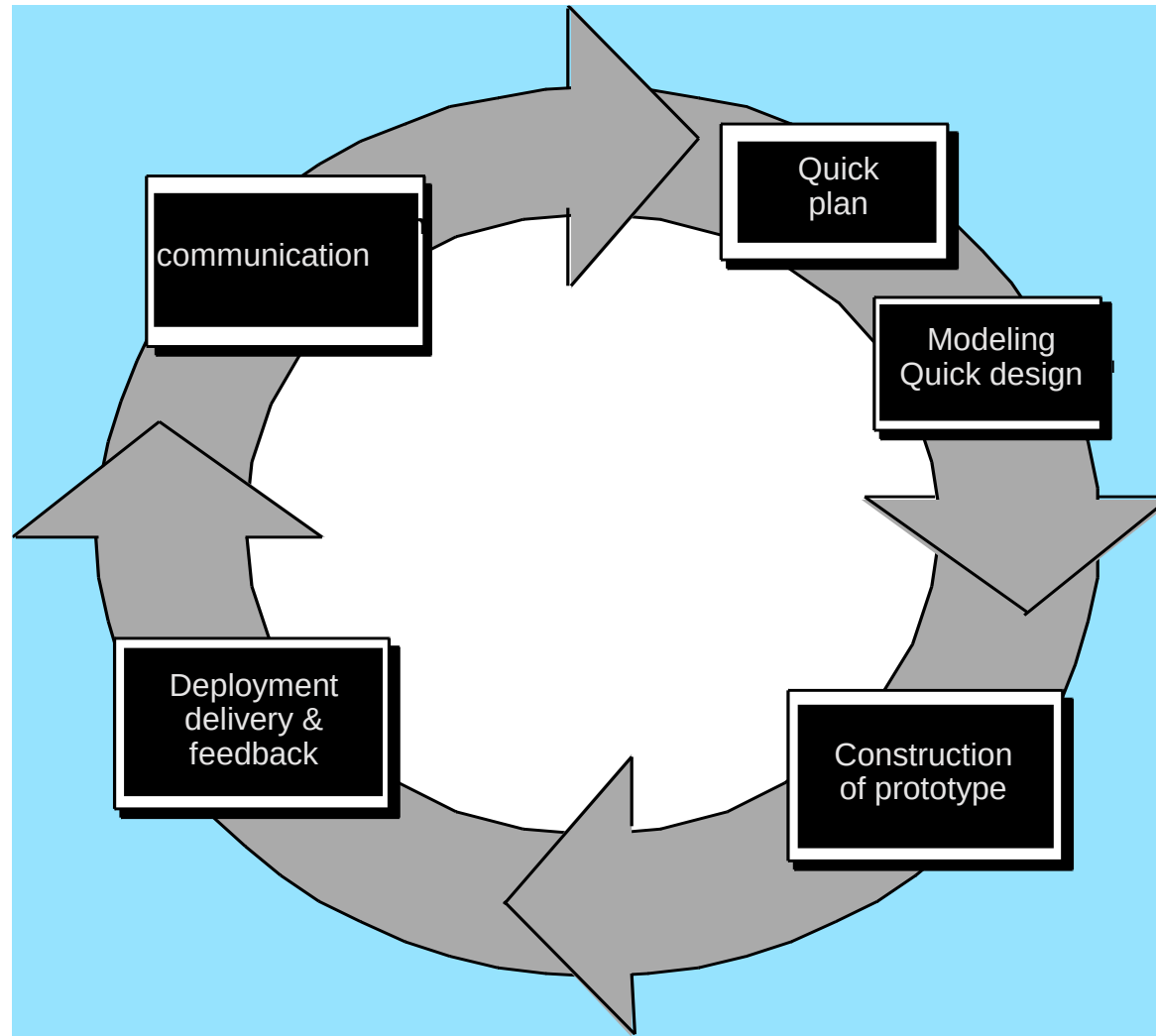
The Incremental Model



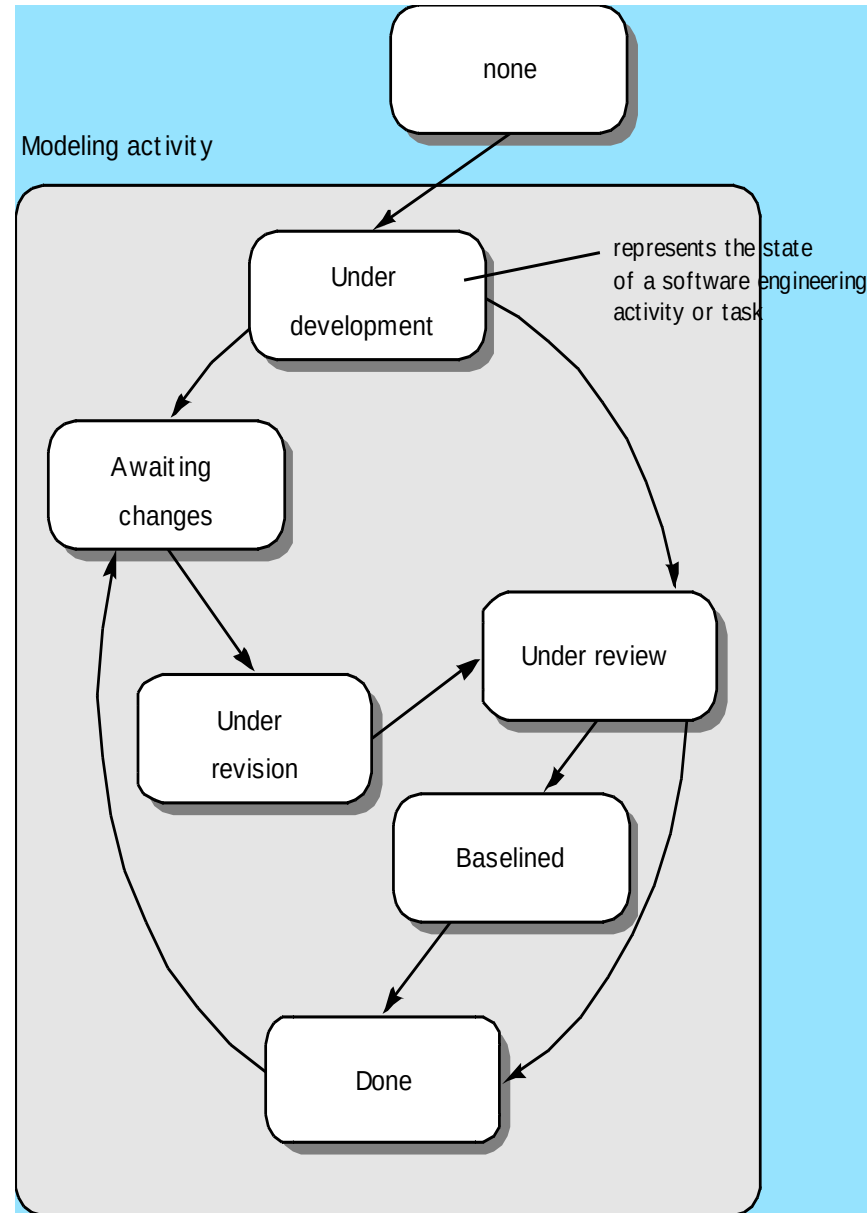
Fountain Model: Object-Oriented



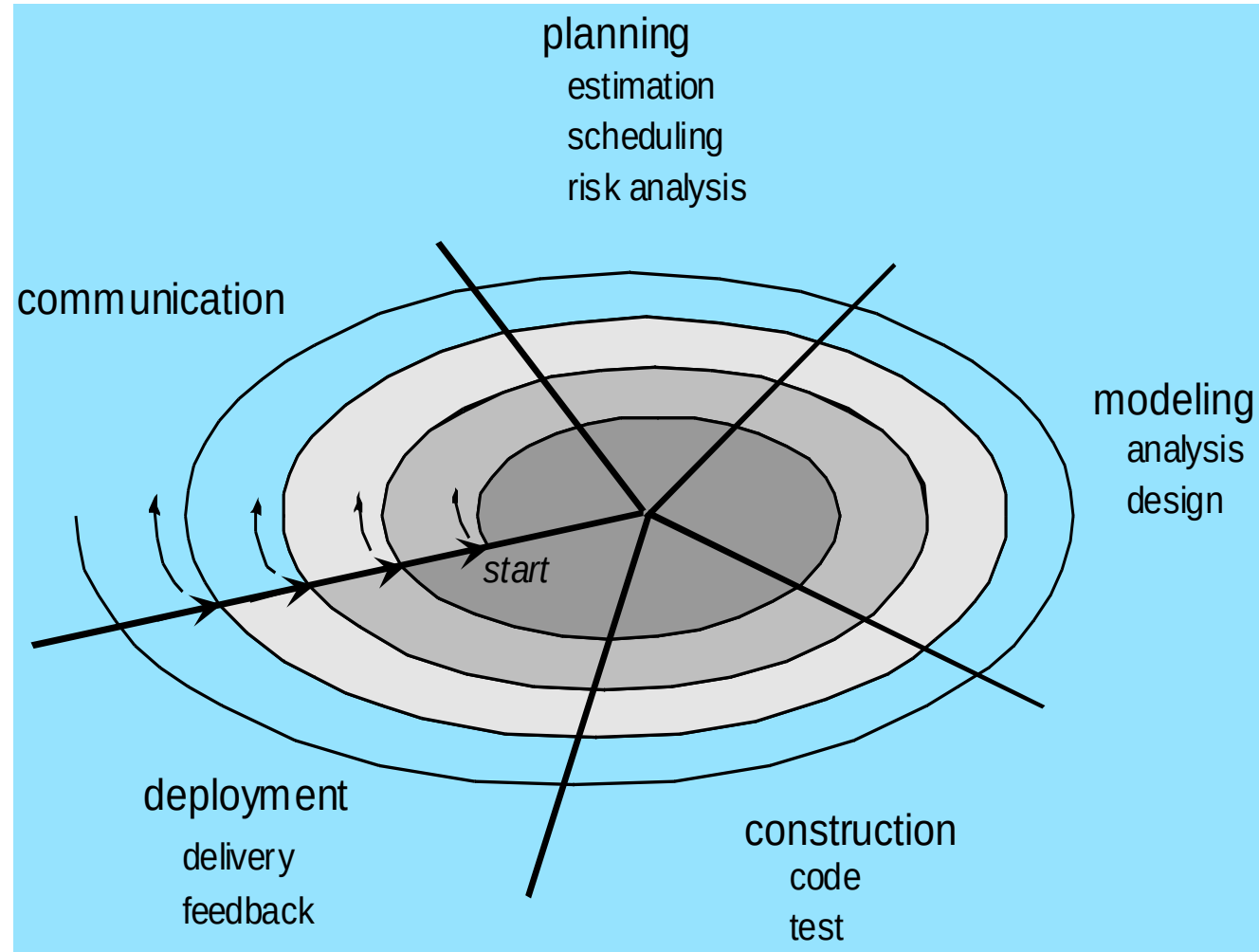
Evolutionary Models: Prototyping



Evolutionary Models: Concurrent



Evolutionary Models: The Spiral



Scrum

Scrum 是一个用于开发和维护复杂产品的框架，是一个增量的、迭代的开发过程，是全球运用最广泛的敏捷管理框架。



- SCRUM框架包括3个角色、3个工件、5个事件、5个价值

3个角色

- 产品负责人 (Product Owner)
- Scrum Master
- 开发团队

3个工件

- 产品Backlog (Product Backlog)
- SprintBacklog
- 产品增量 (Increment)

5个事件

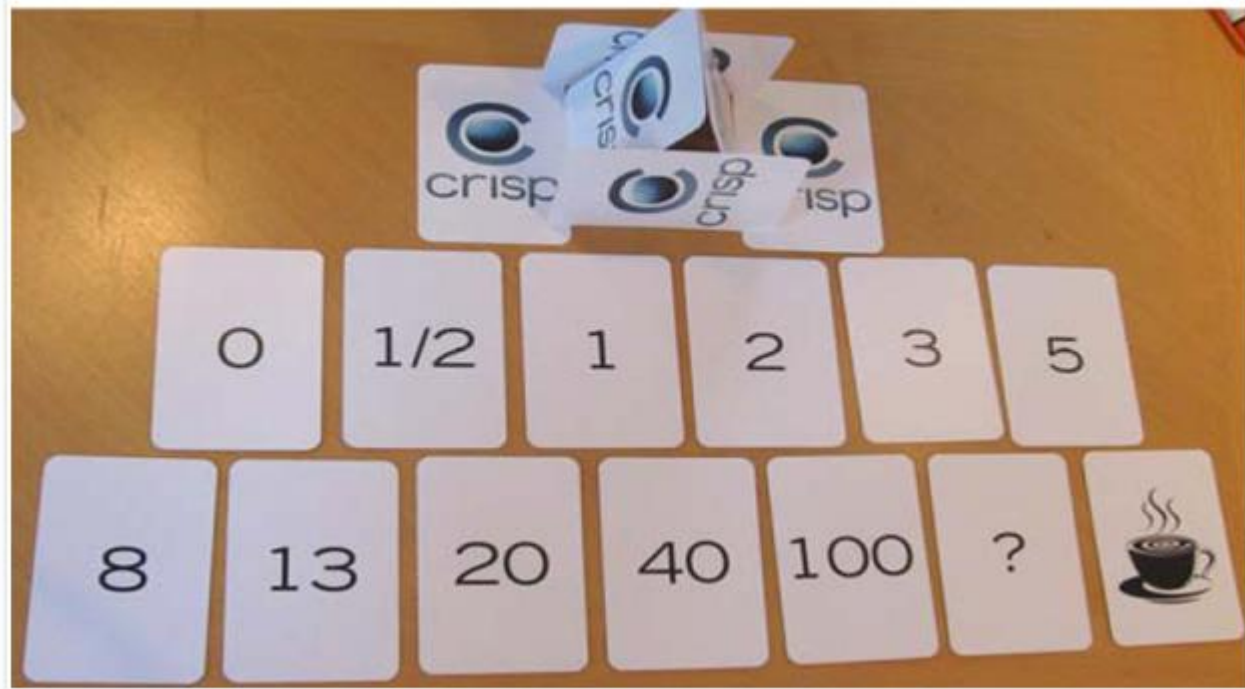
- Sprint（Sprint本身是一个事件，包括了如下4个事件）
 - Sprint计划会议（Sprint Planning Meeting）：计划纸牌
 - 每日站会（Daily Scrum Meeting）
 - Sprint评审会议（Sprint Review Meeting）
 - Sprint回顾会议（Sprint Retrospective Meeting）

5个价值

- 承诺 – 愿意对目标做出承诺
- 专注– 把你的心思和能力都用到你承诺的工作上去
- 开放– Scrum 把项目中的一切开放给每个人看
- 尊重– 每个人都有他独特的背景和经验
- 勇气– 有勇气做出承诺，履行承诺，接受别人的尊重

Scrum

计划纸牌



每日站立会议



SCRUM理论基础

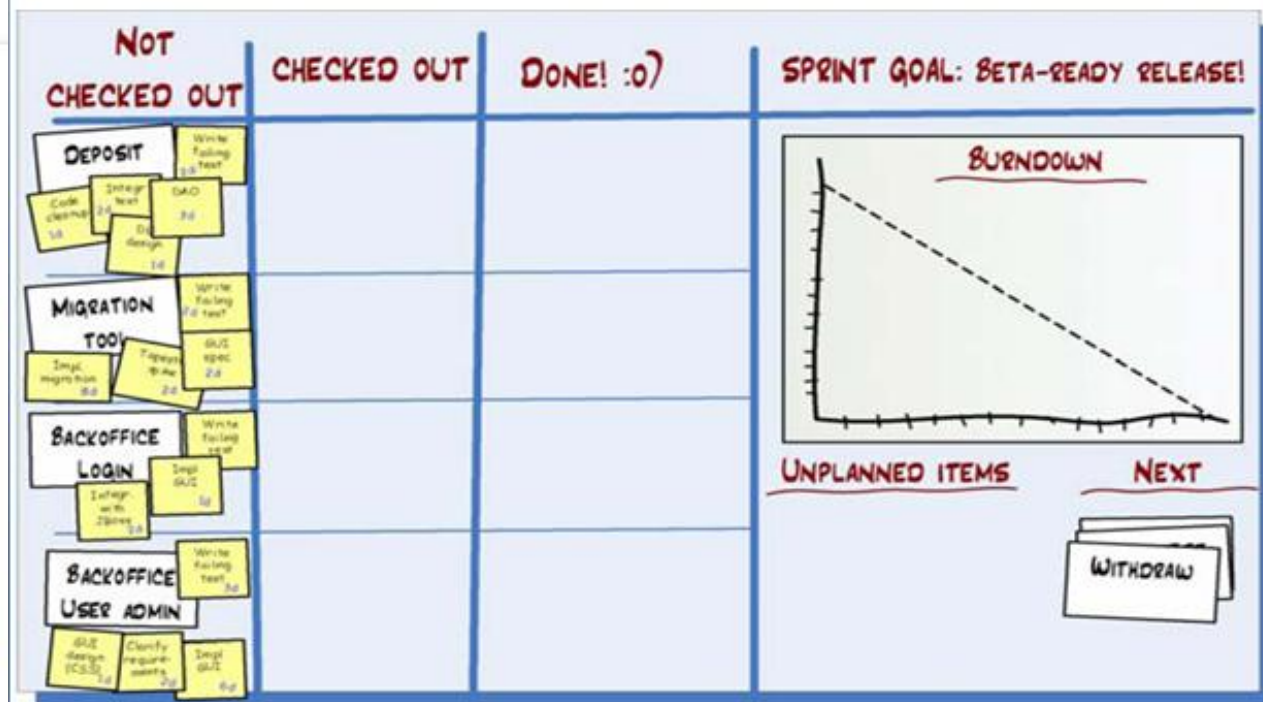
- Scrum以经验性过程控制理论（经验主义）做为理论基础的过程。经验主义主张知识源于经验, 以及基于已知的东西做决定。Scrum采用迭代、增量的方法来优化可预见性并控制风险。
- Scrum 的三大支柱支撑起每个经验性过程控制的实现: 透明性、检验和适应。
 - **透明性**是指, 在软件开发过程的各个环节保持高度的可见性, 影响交付成果的各个方面对于参与交付的所有人、管理生产结果的人保持透明。管理生产成果的人不仅要能够看到过程的这些方面, 而且必须理解他们看到的内容。也就是说, 当某个人在检验一个过程, 并确信某一个任务已经完成时, 这个完成必须等同于他们对完成的定义。

Scrum

任务看板



任务看板



可视化实时协作看板

- Leangoo（中文名：领歌）诞生于2014年，是一款基于看板的[项目协作软件](#)，目前已为数百万企业提供项目协作支持。通过Leangoo软件，我们可以非常方便的管理项目，实现需求、任务、问题和文档的管理和协作，并随时随地的跟踪团队工作进展。完美支持看板方法和[Scrum敏捷](#)开发。
- Leangoo支持灵活的自定义看板结构，创建列表即可轻松定义工作流程，创建泳道可以对任务进行分组对应，拖动任务以体现工作进展。实时同步看板信息，让团队工作情况一目了然，让团队成员即时协作，效率倍增。
- Leangoo基于Scrum框架提供了一系列的流程和模板，可以帮助团队快速启动Scrum敏捷开发，开展基于敏捷的[产品路线图规划](#)，[需求管理](#)、[迭代任务协作](#)和[缺陷跟踪](#)。

Scrum-Leangoo领歌



产品Backlog

敏捷需求管理



产品路线图

敏捷产品规划



Scrum任务板

基于看板的高效协作



故事地图

用户故事地图



用户故事

敏捷的需求表达方式



验收条件

定义需求的验收标准



缺陷管理

基于看板管理缺陷



持续集成

打通看板和DevOps平台



燃尽图

跟踪团队的开发进度



团队速度

跟踪团队的开发效率



任务分布

团队成员的工作分布



故事点估算

基于敏捷的工作量估算

Leangoo支持灵活的自定义看板结构，创建列表即可轻松定义工作流程，创建泳道可以对任务进行分组对应，拖动任务以体现工作进展。实时同步看板信息，让团队工作情况一目了然，让团队成员即时协作，效率倍增。

Leangoo

The screenshot displays the Leangoo web application interface. At the top, there's a navigation bar with the Leangoo logo, a search bar, and links for '产品' (Product), '应用场景' (Application Scenarios), '价格' (Price), '帮助' (Help), '移动端' (Mobile), and '关于我们' (About Us). There are also buttons for '登录' (Login) and '免费使用' (Free Use). Below the navigation bar, a blue banner highlights the application's features: 'Leangoo支持灵活的自定义看板结构，创建列表即可轻松定义工作流程，创建泳道可以对任务进行分组对应，拖动任务以体现工作进展。实时同步看板信息，让团队工作情况一目了然，让团队成员即时协作，效率倍增。'

The main interface shows a project named '第一个迭代' (First Iteration) for the period '7月15日 - 7月29日' (July 15 - July 29) under the '网上书城项目' (Online Bookstore Project). The project is set to '项目成员可见' (Visible to project members). A sidebar on the left contains a '需求(2)' (Requirements) section with a card titled '作为消费者，我希望可以在书店以便我能第一时间...' and a '添加卡片' (Add Card) button. The main area shows a '待办' (To Do) section with a task card titled '创建一个搜索输入框' (Create a search input box). The card has a member '成员' (Member) and a tag '标签' (Tag) '前端开发' (Frontend Development). Below the title, there's a section for '编辑描述' (Edit Description) with the text '输入框占位文字默认显示为最近热搜的书籍。' (Input box placeholder text defaults to the most popular books recently searched). There's a '检查项' (Checklist) section with three items: '输入框内默认占位文字从后台加载' (Default placeholder text in input box loaded from backend), '搜索时相似内容展示' (Display similar content when searching), and '搜索内容展示区域' (Display search content area). Below the checklist is a text input field for '输入检查项...' (Input check item...) with a hint 'shift+enter换行，回车提交' (shift+enter for new line, enter for submit). There's also a '评论' (Comments) section with a text input field for '输入评论...' (Input comment...) with a hint 'shift+enter换行，回车提交，可@其他看板成员' (shift+enter for new line, enter for submit, can @ other board members). On the right side of the task card, there's a '添加' (Add) section with buttons for '成员' (Member), '标签' (Tag), '检查项' (Checklist), '工作量' (Workload), '附件' (Attachment), '开始时间' (Start time), '截止时间' (End time), and '引用' (Cite). Below this is an '操作' (Actions) section with buttons for '复制' (Copy), '移动' (Move), '轨迹' (Track), '分享' (Share), and '删除' (Delete). A '工作量估算' (Workload Estimation) modal is open, showing a list of values: 0, 0.5, 1, 2, 3, 5, 8, 13, 20, 40, and a text input for '输入自定义值(支持数字和1位小数)' (Input custom value (supports numbers and 1 decimal place)).

Leangoo支持灵活的自定义看板结构，创建列表即可轻松定义工作流程，创建泳道可以对任务进行分组对应，拖动任务以体现工作进展。实时同步看板信息，让团队工作情况一目了然，让团队成员即时协作，效率倍增。

The screenshot displays the Leangoo Kanban board interface for a project named "网上书城项目" (Online Bookstore Project). The interface is organized into four main columns: "需求" (Requirements), "待办" (To Do), "进行中" (In Progress), and "已完成" (Completed). Each column contains task cards with details such as assignees, due dates, and progress indicators.

需求 (2):

- 作为消费者，我希望可以在书城首页查找书籍，以便我能第一时间找到我想要的书。

待办 (6):

- 前端开发**
 - 创建一个搜索输入框 (Assignee: [Avatar], Due: 今天09:00, Progress: 2/3)
 - 后端开发**
 - 查询搜索内容并返回前台 (Assignee: [Avatar], Due: 明天截止, Progress: 3/5)
 - 测试**
 - 测试浏览器兼容性 (Assignee: [Avatar], Progress: 0/6)

进行中 (0):

- 添加卡片

已完成 (0):

- 添加卡片

Bottom Section:

- 需求 (2):**
 - 作为消费者，我希望可以通过微信登录书城，以便我可以避免每次都输入手机号或邮箱带来的麻烦。
- 待办 (6):**
 - 前端开发**
 - 登录页面添加微信登录按钮 (Assignee: [Avatar], Due: 7月24日开始, Progress: 1)
 - 后台处理微信账号和商城账号绑定
 - 测试不同浏览器是否正常跳转登录
- 进行中 (0):**
 - 添加卡片
- 已完成 (0):**
 - 添加卡片

实训1

- 使用Leangoo进行项目管理
- <https://www.leangoo.com/>
- <https://docs.leangoo.com/>

SCRUM理论基础

- Scrum 的三大支柱支撑起每个经验性过程控制的实现:透明性、检验和适应。
 - 开发过程中的各方面必须做到足够频繁地检验，确保能够及时发现过程中的重大偏差。在确定检验频率时，需要考虑到检验会引起所有过程发生变化。当规定的检验频率超出了过程检验所能容许的程度，那么就会出现问題。幸运的是，软件开发并不会出现这种情况。另一个因素就是检验工作成果人员的技能水平和积极性。
 - 如果检验人员检验的时候发现过程中的一个或多个方面不满足验收标准，并且最终产品是不合格的，那么便需要对过程或是材料进行适应性调整。调整工作必须尽快实施，以减少进一步的偏差。

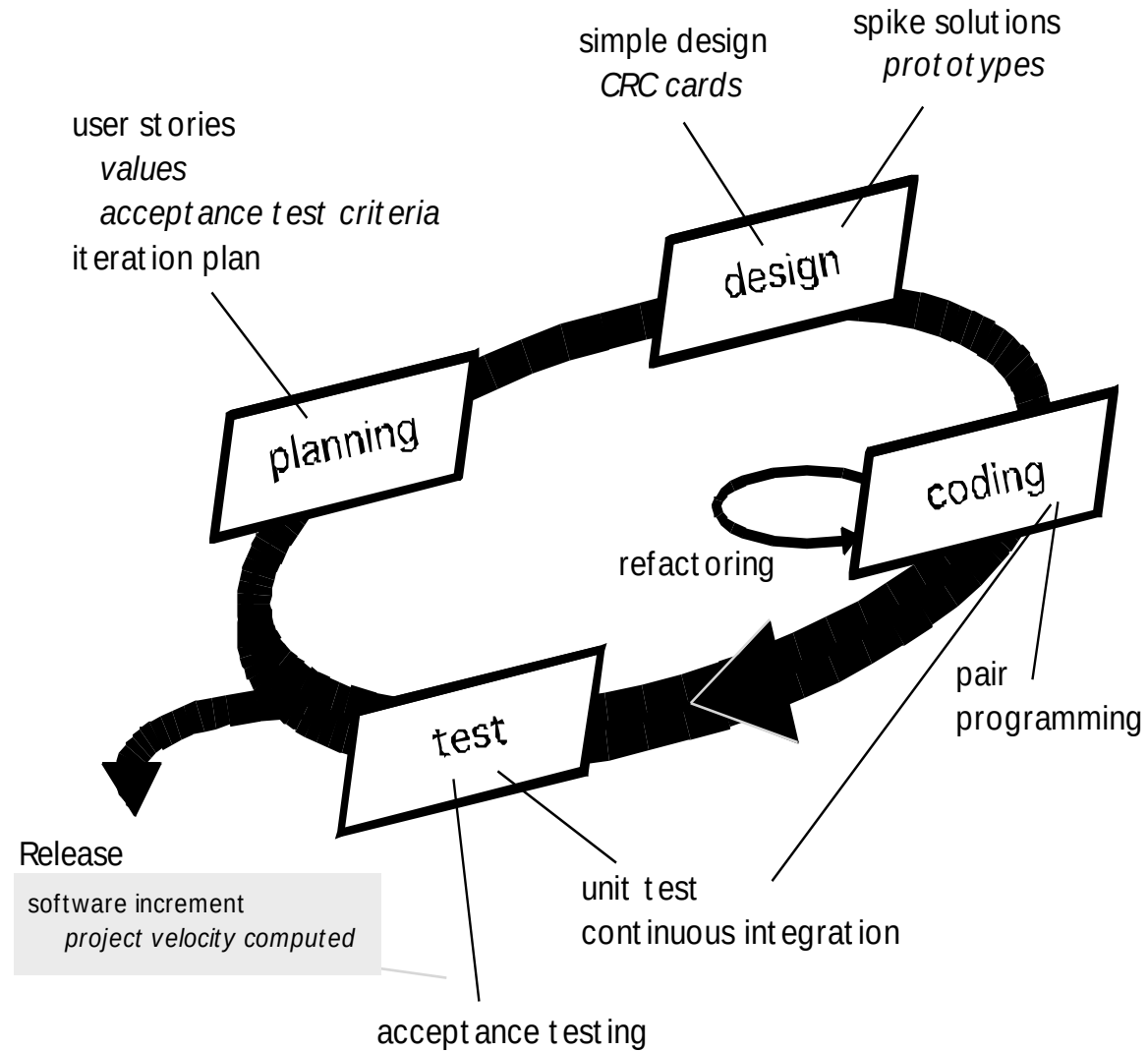
Extreme Programming (XP)

- The most widely used agile process, originally proposed by Kent Beck
- XP Planning
 - Begins with the creation of “user stories”
 - Agile team assesses each story and assigns a cost
 - Stories are grouped to for a deliverable increment
 - A commitment is made on delivery date
 - After the first increment “project velocity”（速度） is used to help define subsequent delivery dates for other increments

Extreme Programming (XP)

- XP Design
 - Follows the **KIS principle**
 - Encourage the use of **CRC cards**
 - For difficult design problems, suggests the creation of “**spike solutions**”—a design prototype
 - Encourages “**refactoring**”—an iterative refinement of the internal program design
- XP Coding
 - Recommends the **construction of a unit test** for a store *before* coding commences
 - Encourages “**pair programming**”
- XP Testing
 - All **unit tests** are executed daily
 - “**Acceptance tests**” are defined by the customer and executed to assess customer visible functionality

Extreme Programming (XP)



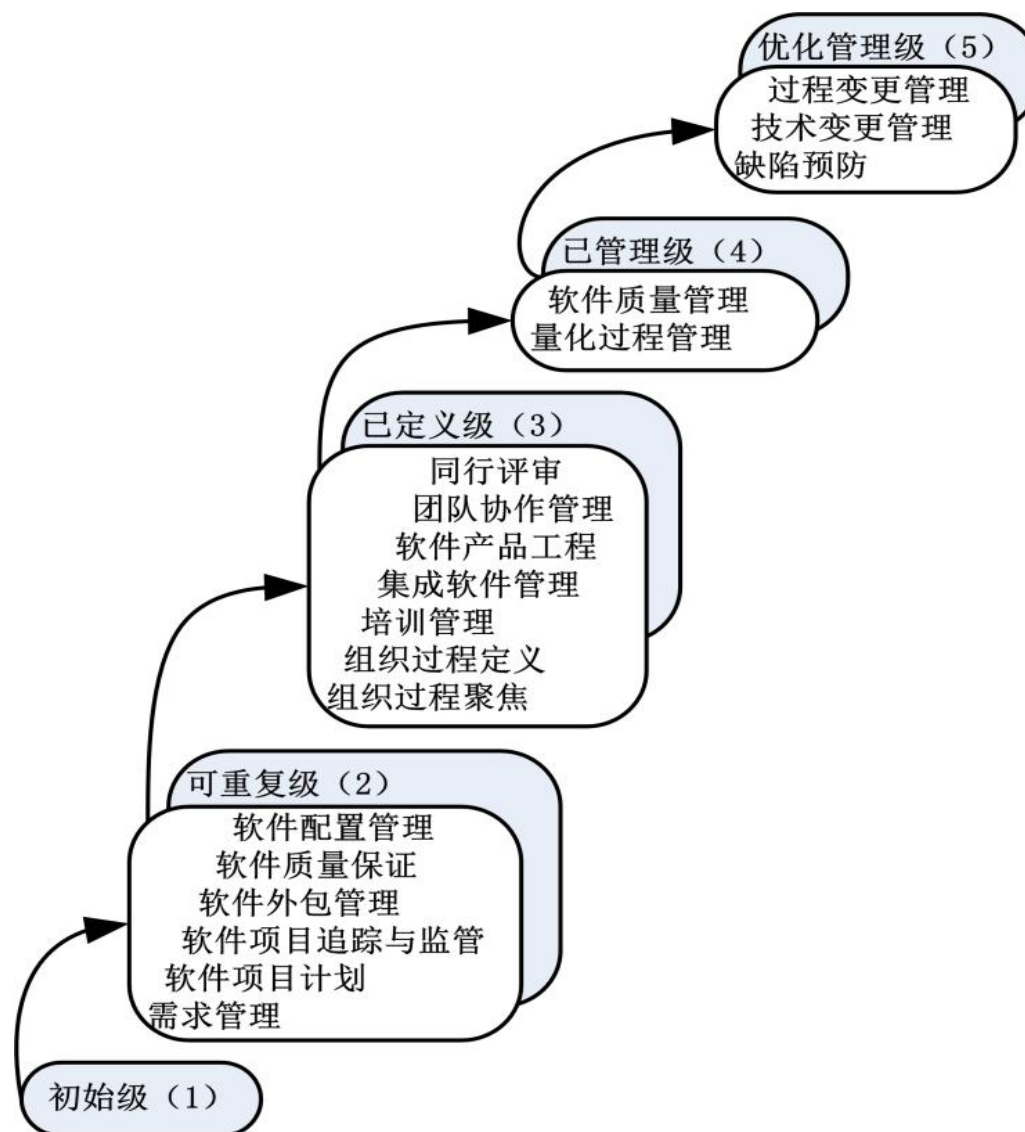
Other Process Models

- **Component based development**—the process to apply when reuse is a development objective
- **Formal methods**—emphasizes the mathematical specification of requirements
- **AOSD**—provides a process and methodological approach for defining, specifying, designing, and constructing *aspects*
- **Unified Process**—a “use-case driven, architecture-centric, iterative and incremental” software process closely aligned with the Unified Modeling Language (UML)

Process Assessment and Improvement

- **Standard CMMI Assessment Method for Process Improvement (SCAMPI)** — provides a five step process assessment model that incorporates five phases: initiating, diagnosing, establishing, acting and learning.
- **CMM-Based Appraisal for Internal Process Improvement (CBA IPI)** — provides a diagnostic technique for assessing the relative maturity of a software organization; uses the SEI CMM as the basis for the assessment [Dun01]
- **SPICE—The SPICE (ISO/IEC15504)** standard defines a set of requirements for software process assessment. The intent of the standard is to assist organizations in developing an objective evaluation of the efficacy of any defined software process. [ISO08]
- **ISO 9001:2000 for Software** — a generic standard that applies to any organization that wants to improve the overall quality of the products, systems, or services that it provides. Therefore, the standard is directly applicable to software organizations and companies. [Ant06]

Key Process Areas



Personal Software Process (PSP)

- **Planning.** This activity isolates requirements and develops both size and resource estimates. In addition, a defect estimate (the number of defects projected for the work) is made. All metrics are recorded on worksheets or templates. Finally, development tasks are identified and a project schedule is created.
- **High-level design.** External specifications for each component to be constructed are developed and a component design is created. Prototypes are built when uncertainty exists. All issues are recorded and tracked.
- **High-level design review.** Formal verification methods are applied to uncover errors in the design. Metrics are maintained for all important tasks and work results.
- **Development.** The component level design is refined and reviewed. Code is generated, reviewed, compiled, and tested. Metrics are maintained for all important tasks and work results.
- **Postmortem** 事后检查. Using the measures and metrics collected (this is a substantial amount of data that should be analyzed statistically), the effectiveness of the process is determined. Measures and metrics should provide guidance for modifying the process to improve its effectiveness.

Team Software Process (TSP)

- Build self-directed teams that plan and track their work, establish goals, and own their processes and plans. These can be pure software teams or integrated product teams (IPT) of **three to about 20** engineers.
- Show managers how to **coach and motivate** their teams and how to help them sustain peak performance.
- Accelerate software process improvement by making **CMM Level 5** behavior normal and expected.
 - The Capability Maturity Model (CMM), a measure of the effectiveness of a software process.
- Provide improvement guidance to high-maturity organizations.
- Facilitate university teaching of industrial-grade team skills.

谢谢!